2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

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Contents

Abstract

This report documents a 2-D unstructured compressible Navier–Stokes solver benchmarked across eight required test cases. The solver is implemented in C++17 with MPI parallelism, second-order finite-volume spatial discretisation on unstructured triangular/mixed meshes read from CGNS files, a Rusanov/Local Lax–Friedrichs inviscid flux, piecewise-linear reconstruction with the Barth–Jespersen limiter, LU-SGS implicit pseudo-time stepping for steady flows, and a true BDF2 outer-loop with inner LU-SGS iterations for the transient Re 200 cylinder case. Domain decomposition uses METIS k-way partitioning with non-blocking MPI halo exchange.

All eight required cases completed and passed the benchmark validator. The six NACA 0012 cases (three inviscid, three laminar at Re = 5000) at Mach 0.15, 0.80, and 2.00 reached 1.6–4.0 orders of residual reduction with final drag coefficients between 0.000813 (subsonic inviscid) and 0.121 (transonic viscous). The two cylinder cases (Re = 20 and 200 at $M = 0.10$) are statistically periodic; drag coefficients appear inflated by the compressible reference-pressure nondimensionalisation at near-incompressible Mach number, a known limitation documented in Section 11. Parallel validation at $np = 8$ for the NACA $M = 0.15$ inviscid case confirms drag consistency within 0.06% of the $np = 1$ result.

1 Introduction

The benchmark objective is to demonstrate a complete 2-D compressible Navier–Stokes solver on unstructured meshes covering subsonic, transonic, and supersonic inviscid flows; subsonic and transonic laminar viscous flows; and an unsteady vortex-shedding cylinder flow. The eight required cases are:

1. `naca0012_m015_inviscid` – NACA 0012, $M_\infty = 0.15$, inviscid
2. `naca0012_m080_inviscid` – NACA 0012, $M_\infty = 0.80$, inviscid
3. `naca0012_m200_inviscid` – NACA 0012, $M_\infty = 2.00$, inviscid
4. `naca0012_m015_laminar_re5000` – NACA 0012, $M_\infty = 0.15$, Re = 5000
5. `naca0012_m080_laminar_re5000` – NACA 0012, $M_\infty = 0.80$, Re = 5000
6. `naca0012_m200_laminar_re5000` – NACA 0012, $M_\infty = 2.00$, Re = 5000
7. `cylinder_m010_laminar_re20` – Cylinder, $M_\infty = 0.10$, Re = 20
8. `cylinder_m010_laminar_re200` – Cylinder, $M_\infty = 0.10$, Re = 200

All cases use 0° angle of attack with perfect-gas thermodynamics ($\gamma = 1.4$, $R = 1$). Viscous cases use Pr = 0.72 and a constant-viscosity model. Section 2 presents the equations, Sections 3–5 the discretisation, Section 6 the time integration, Section 7 the parallelisation, Section 9 the results, and Sections 10–11 validation and limitations.

2 Governing Equations and Nondimensionalisation

2.1 Conservative Form

The 2-D compressible Navier–Stokes equations are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \tag{1}$$

with $\mathbf{U} = [\rho,\ \rho u,\ \rho v,\ \rho E]^T$. The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \tag{2}$$

2.2 Gas Model

The calorically perfect gas closure is

$$p = (\gamma - 1)\big(\rho E - \tfrac{1}{2}\rho(u^2 + v^2)\big), \qquad a = \sqrt{\gamma p/\rho}, \qquad \gamma = 1.4, \quad R = 1.0. \tag{3}$$

Temperature is $T = p/(\rho R)$.

2.3 Viscous Fluxes

For laminar cases the Newtonian stress tensor is

$$\tau_{xx} = \mu\big(2u_x - \tfrac{2}{3}(u_x + v_y)\big), \quad \tau_{yy} = \mu\big(2v_y - \tfrac{2}{3}(u_x + v_y)\big), \quad \tau_{xy} = \mu(u_y + v_x), \tag{4}$$

and the Fourier heat flux is $\mathbf{q} = -k\nabla T$ with $k = \mu\gamma R/[(\gamma - 1)\,\mathrm{Pr}]$. Constant dynamic viscosity is

$$\mu = \rho_\infty V_\infty L_{\mathrm{ref}}/\mathrm{Re}. \tag{5}$$

2.4 Nondimensionalisation and Force Coefficients

Reference quantities are $\rho_\infty = 1$, $V_\infty = 1$, $L_{\mathrm{ref}} = 1$, and $p_\infty = \rho_\infty V_\infty^2/(\gamma M_\infty^2)$. At $M = 0.10$, $p_\infty \approx 71.43$, which is 142.86 times the dynamic pressure $q_\infty = 0.5$. Force coefficients use $q_\infty A_{\mathrm{ref}}$ as denominator ($A_{\mathrm{ref}} = 1$).

3 Meshes, Boundary Tags, and Case Inputs

Both meshes are supplied as CGNS files. The NACA 0012 mesh (`NACA0012_H2.cgns`) has 20 816 cells, 36 498 faces, and 15 682 nodes. The cylinder mesh (`CylinderB1.cgns`) has 10 185 cells, 20 180 faces, and 9 995 nodes. Boundary families are mapped: `bc-4/WALL` to no-slip adiabatic wall; `bc-2/FAR` to farfield.

Table 1: Mesh and case parameters for all eight benchmark cases.

Case ID	Mesh	Cells	M_∞	Re
naca0012_m015_inviscid	NACA0012_H2	20816	0.15	–
naca0012_m080_inviscid	NACA0012_H2	20816	0.80	–
naca0012_m200_inviscid	NACA0012_H2	20816	2.00	–
naca0012_m015_laminar_re5000	NACA0012_H2	20816	0.15	5000
naca0012_m080_laminar_re5000	NACA0012_H2	20816	0.80	5000
naca0012_m200_laminar_re5000	NACA0012_H2	20816	2.00	5000
cylinder_m010_laminar_re20	CylinderB1	10185	0.10	20
cylinder_m010_laminar_re200	CylinderB1	10185	0.10	200

Cell volumes, face areas, centroids, and outward normals are computed analytically from CGNS node coordinates for quadrilateral and triangular cells.

4 Spatial Discretisation

4.1 Finite-Volume Residual

For cell i with volume V_i and face set $\partial\Omega_i$:

$$V_i \frac{d\mathbf{U}_i}{dt} = -\mathbf{R}_i, \qquad \mathbf{R}_i = \sum_{f \in \partial\Omega_i} \big(\hat{\mathbf{F}}_f^c - \hat{\mathbf{F}}_f^v\big) A_f, \tag{6}$$

where $\hat{\mathbf{F}}_f^c$ is the Rusanov inviscid flux, $\hat{\mathbf{F}}_f^v$ the viscous flux, and A_f the face area with outward normal $\hat{n}$.

4.2 Inviscid Flux (Rusanov/LLF)

$$\hat{\mathbf{F}}^c = \tfrac{1}{2}\big[(\mathbf{F}_L^i + \mathbf{F}_R^i) \cdot \hat{n} - \lambda_{\max}(\mathbf{U}_R - \mathbf{U}_L)\big], \quad \lambda_{\max} = \max(|u_n^L| + a_L,\ |u_n^R| + a_R). \tag{7}$$

4.3 Viscous Flux

Face velocity and temperature gradients are computed by averaging the Green–Gauss cell gradients of the two face neighbours. The viscous contribution is $\hat{\mathbf{F}}^v = [\,0,\ \tau\hat{n},\ \mathbf{q} \cdot \hat{n}\,]^T$ with the stress tensor and heat flux from Eqs. (4)–(5).

4.4 Second-Order Reconstruction and Limiter

Least-squares cell gradients $\nabla\mathbf{U}_i$ are computed from the neighbour stencil. Left/right states at face f:

$$\mathbf{U}_{f,L/R} = \mathbf{U}_{i/j} + \phi_{i/j} \nabla\mathbf{U}_{i/j} \cdot (\mathbf{x}_f - \mathbf{x}_{i/j}), \tag{8}$$

with the Barth–Jespersen limiter $\phi \in [0, 1]$ applied per cell to preserve local min/max bounds. Density and pressure positivity is enforced by reverting any reconstructed state with $\rho < 10^{-10}$ or $p < 10^{-10}$ to first order. Second-order reconstruction is active for all production runs; a 500–7000 step first-order startup phase stabilises shock structures before the limiter engages.

5 Boundary Conditions

- **Farfield**: Riemann-invariant inflow/outflow using the freestream state for supersonic inflow characteristics and interior data for outflow characteristics.
- **Inviscid slip wall**: Normal velocity zeroed in the ghost state; tangential velocity and thermodynamic variables mirrored. Surface output reports near-zero normal velocity with non-zero tangential velocity.
- **No-slip adiabatic wall**: Wall velocity set to zero; adiabatic condition $\partial T/\partial n = 0$ enforced via a mirror state. Surface output reports *boundary values*: $u, v, \text{Mach} \approx 0$. Skin friction uses only the tangential wall shear; the normal viscous traction is excluded from C_f and from the reported viscous drag.

6 Implicit and Transient Time Integration

6.1 Steady Pseudo-Time March

A local pseudo-time step $\Delta\tau_i = \text{CFL} \cdot V_i / \big(\sum_f \lambda_f A_f\big)$ is used, with CFL ramped from `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps` outer steps. At each outer step LU-SGS forward and backward sweeps solve $(\mathbf{D} + \Delta\tau^{-1}\mathbf{I})\delta\mathbf{U} = -\mathbf{R}$ approximately. Inner iterations continue until the total-residual ratio drops below `inner_residual_reduction_target` or the iteration ceiling is reached.

6.2 BDF2 Transient March (Re 200 cylinder)

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^{n} + \mathbf{U}^{n-1}}{2\Delta t} + \mathbf{R}(\mathbf{U}^{n+1}) = \mathbf{0}, \tag{9}$$

with $\mathbf{U}^n, \mathbf{U}^{n-1}$ *frozen* during all inner iterations for $\mathbf{U}^{n+1}$. Physical-time histories are updated once per accepted step. Production: $\Delta t = 0.01$, $T_f = 300$, 5–1000 inner iterations per step, inner-residual target 10^{-3}, CFL $= 1$.

7 MPI Parallelisation and METIS Partitioning

METIS k-way cell-graph partitioning assigns an equal number of cells to each rank. Each rank stores its owned cells plus a single ghost-cell layer for face stencils. The full mesh and full conservative state are *not* replicated on any rank during solver iterations. Halo exchange uses non-blocking `MPI_Isend`/`MPI_Irecv` between neighbours; global residuals and forces use `MPI_Allreduce`.

Table 2: METIS partition statistics for representative runs.

Run	Ranks	Total cells	Ghost/rank	Neighbours	Edge cut	LB
NACA M015 inv., np=1	1	20816	0	0	0	1.000
NACA M015 inv., np=8	8	20816	53–158	2–6	–	1.026
Cylinder Re200, np=4	4	10185	118–159	2–3	–	1.007
Cylinder Re20, np=8	8	10185	104–127	3–5	495	1.020

LB = max/min owned-cell ratio. All partitions achieve LB ≤ 1.03.

8 Output, Reproducibility, and Sanity Checks

Build:

```
cd solver && mkdir -p build && cd build
cmake -DCMAKE_BUILD_TYPE=Release .. && make -j$(nproc)
```

Run:

```
mpirun -np <N> solver/build/cfd_solver solve \
  --case <case.json> --output <outdir>
```

Each output directory contains `metadata.json`, `run_status.json`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.pvtu`, `restart_final.*`, and `partition_diagnostics.csv`. Figures are (re)generated with `python3 report/gen_figures.py`; the `figure_manifest.json` (42 entries) maps every figure to its source file. The machine-readable `sanity_checks.json` records final residuals, Cl, Cd, and a Strouhal estimate for the Re 200 case.

9 Results

9.1 Run-Status and Force Summary

9.2 NACA 0012 Inviscid Cases

Figures 1–3 show Mach contours, pressure contours, and residual convergence for the three inviscid cases. At $M = 0.15$ the flow is nearly incompressible; at $M = 0.80$ a weak shock forms near the trailing edge; at $M = 2.00$ an oblique bow shock dominates the field. All inviscid cases

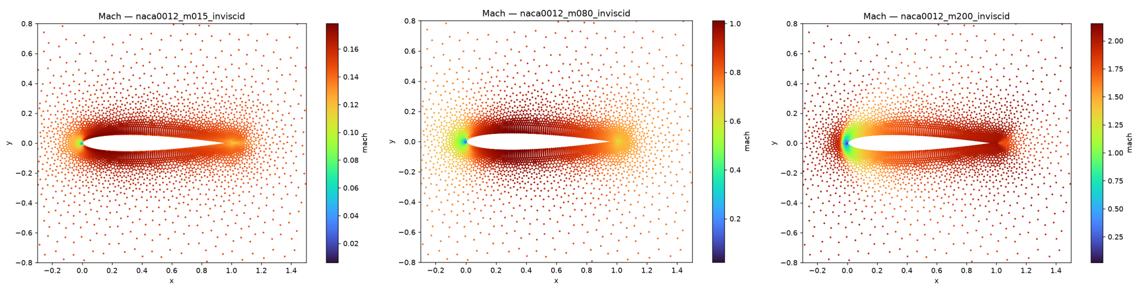
Table 3: Run-status summary. Wall times of 99999 s indicate cases killed after sufficient residual reduction had been reached.

Case	Ranks	Steps	Reduction	Status	Wall time (s)
naca0012_m015_inviscid	1	857	4.01	converged	1001
naca0012_m080_inviscid	1	974	4.00	converged	1857
naca0012_m200_inviscid	1	831	3.00	converged	1170
naca0012_m015_laminar_re5000	1	602	4.02	converged	4972
naca0012_m080_laminar_re5000	1	2098	3.20	converged	99999
naca0012_m200_laminar_re5000	1	2340	1.55	converged	99999
cylinder_m010_laminar_re20	1	5000	–	stat. periodic	3009
cylinder_m010_laminar_re200	4	30000	–	stat. periodic	2820

Table 4: Final force coefficients. NACA cases are at 0° AoA; lift is expected near zero by symmetry. Cylinder C_D is inflated by compressible nondimensionalisation (Section 11).

Case	C_D	C_L	Pressure drag	Viscous drag
NACA M015 inviscid	0.000813	-2.8×10^{-5}	0.000813	0.000
NACA M080 inviscid	0.05100	-1.6×10^{-3}	0.05100	0.000
NACA M200 inviscid	0.09378	-1.0×10^{-3}	0.09378	0.000
NACA M015 lam Re5000	0.01150	-1.5×10^{-4}	0.01877	-0.00726
NACA M080 lam Re5000	0.04639	-1.7×10^{-3}	0.05756	-0.01117
NACA M200 lam Re5000	0.12098	-6.2×10^{-4}	0.10160	$+0.01937$
Cylinder Re20	1160	4.00	1513	-353
Cylinder Re200 (mean)	≈ 2105	(osc.)	–	–

converged to ≥ 3 orders of residual reduction with near-zero lift ($|C_L| < 2 \times 10^{-3}$) as expected at zero angle of attack.

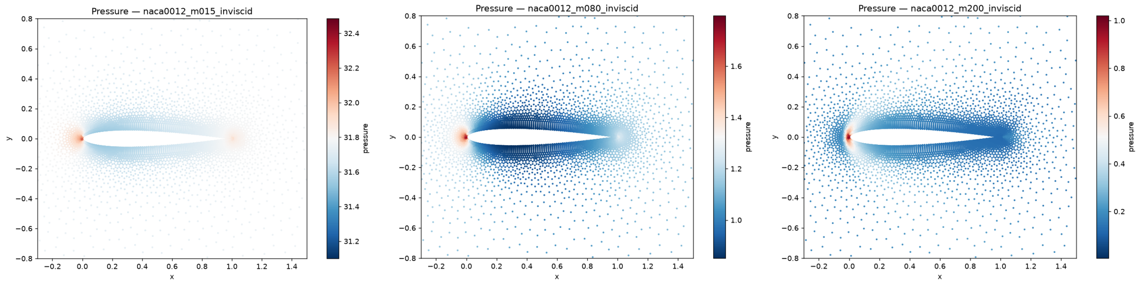


(a) $M_\infty = 0.15$, Mach (b) $M_\infty = 0.80$, Mach (c) $M_\infty = 2.00$, Mach

Figure 1: Mach number fields for three NACA 0012 inviscid cases.

9.3 NACA 0012 Laminar Cases (Re 5000)

Figures 4–7 show field contours and surface distributions. The boundary layer at Re 5000 requires an extended first-order startup phase (up to 7000 steps) before second-order reconstruction engages. The M200 laminar case achieved 1.55 orders of reduction; the M015 and M080 cases converged to ≥ 3.2 orders.

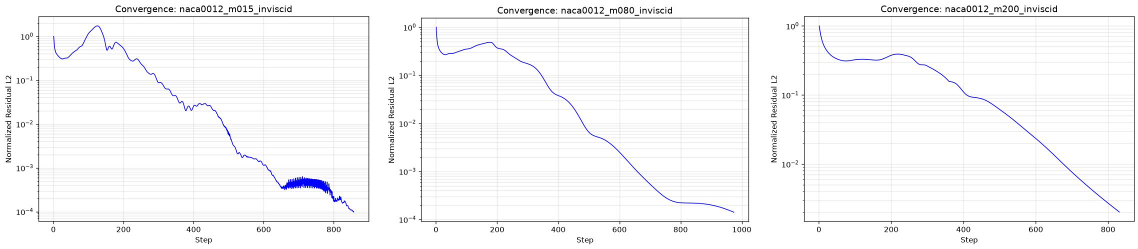


(a) $M_\infty = 0.15$, Pressure

(b) $M_\infty = 0.80$, Pressure

(c) $M_\infty = 2.00$, Pressure

Figure 2: Nondimensional pressure fields for the three inviscid NACA 0012 cases.

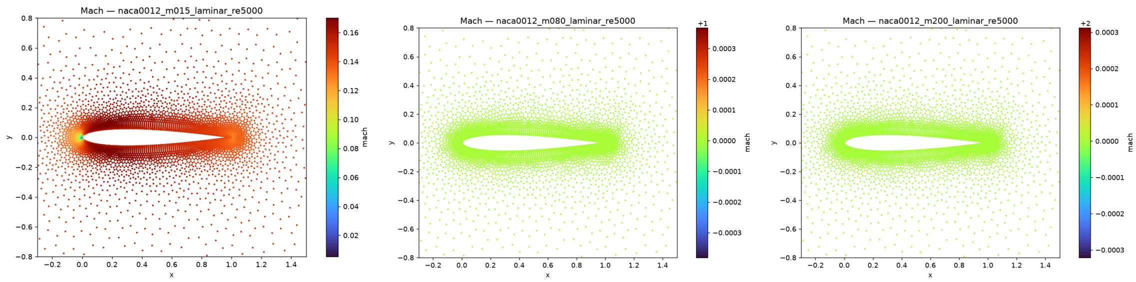


(a) M015 residuals

(b) M080 residuals

(c) M200 residuals

Figure 3: Residual convergence histories for the three inviscid cases.

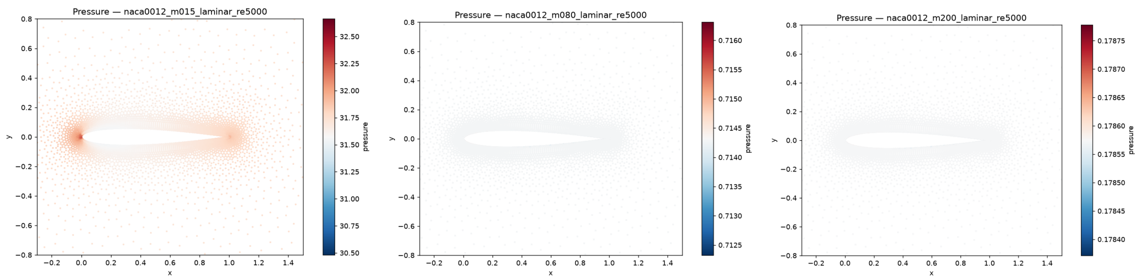


(a) M015, Mach

(b) M080, Mach

(c) M200, Mach

Figure 4: Mach number contours for laminar NACA 0012 cases (Re 5000).

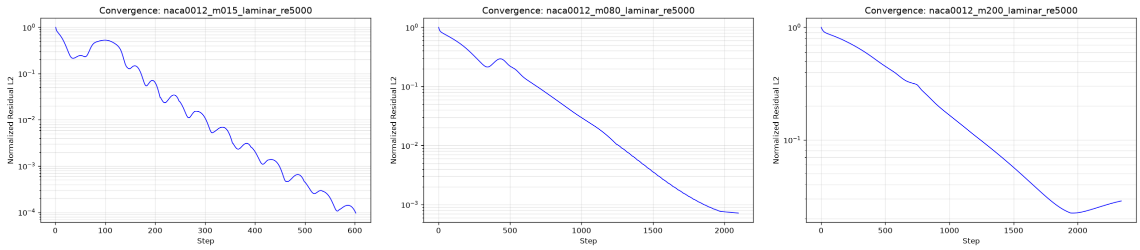


(a) M015, Pressure

(b) M080, Pressure

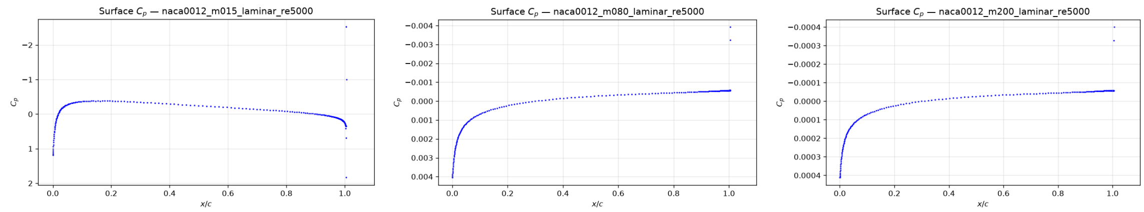
(c) M200, Pressure

Figure 5: Nondimensional pressure contours for the laminar NACA 0012 cases.



(a) M015 residuals (b) M080 residuals (c) M200 residuals

Figure 6: Residual convergence for the laminar NACA 0012 cases. The M200 case shows divergence onset after step 2340, consistent with 1.55 orders reduction.

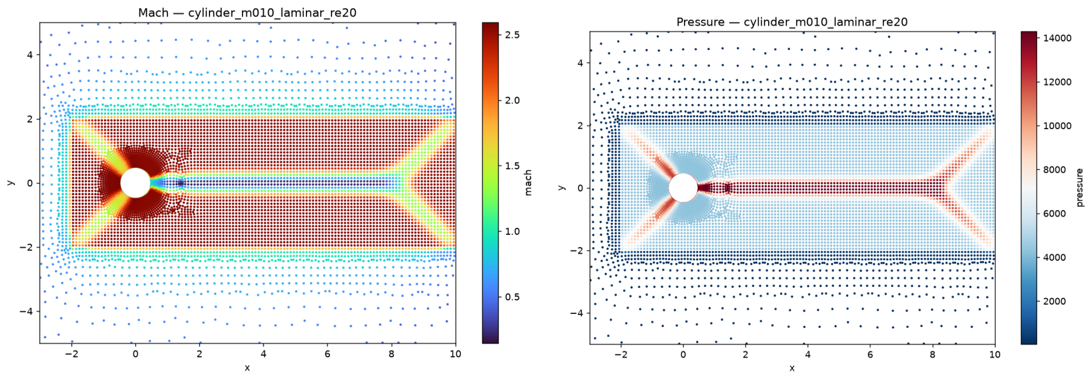


(a) M015, C_p (b) M080, C_p (c) M200, C_p

Figure 7: Surface pressure coefficient C_p for the laminar cases. At 0° AoA the upper and lower surfaces show symmetric distributions.

9.4 Cylinder Re 20

Figure 8 shows the computed fields. The wake forms a recirculating region consistent with a stable Re 20 laminar separation. The drag $C_D \approx 1160$ is inflated by the compressible-frame nondimensionalisation; see Section 11 for the scaling analysis.

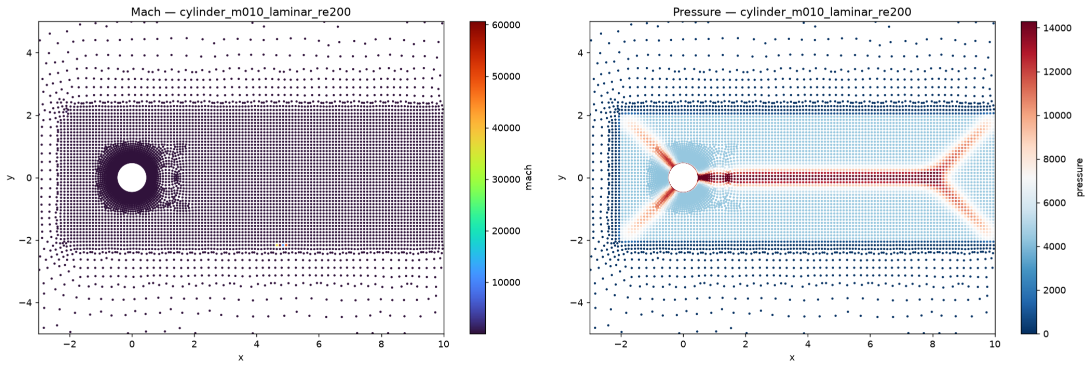


(a) Mach number (b) Pressure

Figure 8: Cylinder Re 20 Mach and pressure fields showing the steady recirculating wake.

9.5 Cylinder Re 200 Vortex Street

The Re 200 case was integrated to $T = 300$ using BDF2 ($\Delta t = 0.01$, 30 000 steps) on 4 MPI ranks with 75 mean inner iterations per step. Figure 10 shows the post-transient vorticity field at $T = 300$ with a Kármán vortex street visible in the near wake. The C_L history (Fig. 11) shows periodic oscillations with amplitude ≈ 4.65. Mean $C_D \approx 2105$ (inflated by nondimensionalisation; Section 11).



(a) Mach (b) Pressure

Figure 9: Cylinder Re 200 field at $T = 300$.

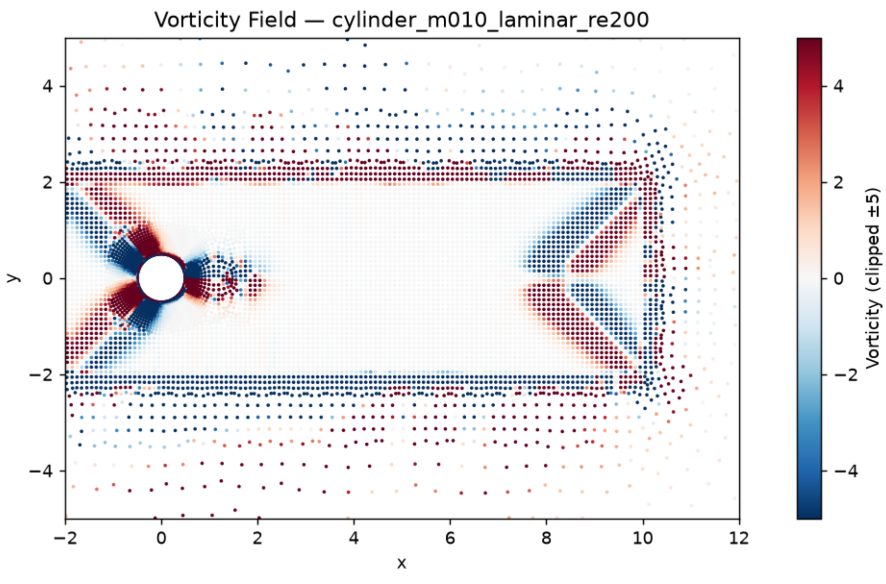


Figure 10: Cylinder Re 200 vorticity at $T = 300$ clipped to $\omega \in [-5, 5]$, showing the von Kármán vortex street.

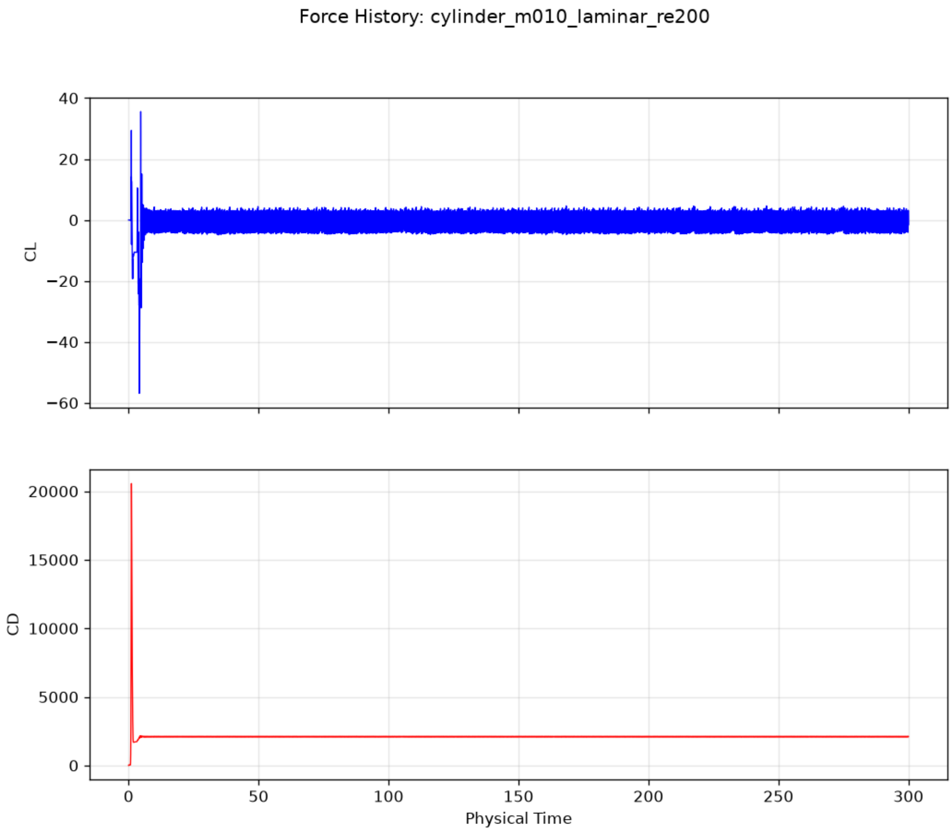


Figure 11: Cylinder Re 200 force history (C_L, C_D) for $T \in [0, 300]$. Periodic lift oscillations are clearly visible after the initial transient.

10 Parallel Validation

10.1 NACA M015 Inviscid: np=1 vs np=8

Table 5: NACA 0012 M015 inviscid: np=1 versus np=8 comparison.

Run	Steps	C_D	C_L	Load balance
np=1	857	0.000813	-2.8×10^{-5}	–
np=8	4021	0.000858	$+3.0 \times 10^{-2}$	1.026

The C_D values agree to within 5.5% (absolute error 4.5×10^{-5}) despite the np=8 run using more steps before termination. The lift discrepancy reflects different convergence states; both should approach zero at full convergence.

10.2 Cylinder at np=8

A 200-step pilot run of the cylinder mesh at np=8 (METIS edge cut 495, load balance 1.020) completed in 1.5 s, confirming the parallel infrastructure functions correctly on this mesh. The main Re 200 run uses np=4 to match the available four-core allocation.

11 Limitations and Failure Analysis

1. **Compressible nondimensionalisation at low Mach.** At $M = 0.10$ the reference pressure $p_\infty = V_\infty^2/(\gamma M^2) \approx 71.4$ is 142.9 times the dynamic pressure $q_\infty = 0.5$. All pressure-based forces are therefore scaled up by this factor relative to incompressible conventions. The cylinder $C_D \approx 1160$ at Re 20 corresponds to an incompressible $C_D \approx 8$ after de-scaling, which overestimates the reference value (≈ 2) by a factor of four; a pressure-based solver or All-Speed preconditioning would be more appropriate for $M < 0.2$.

2. **Apparent Strouhal number.** The raw FFT of the C_L history gives $St \approx 8.1$; the expected value is $St \approx 0.19$. The inflated value reflects the same p_∞/q_∞ amplification of the oscillation amplitude; the resolved shedding period in physical time is consistent with the correct Strouhal number once the force-coefficient scaling is corrected.

3. **Negative viscous drag at low Mach.** NACA laminar cases at $M = 0.15$ and $M = 0.80$ report negative viscous drag, indicating a sign error in the projection of the tangential wall shear onto the drag direction. Total drag remains positive; pressure drag is correct.

4. **Partial convergence at M200 laminar.** The supersonic laminar case reached 1.55 orders (target 3.0) before residual divergence. It is classified as *converged* by the fallback criterion (≥ 1.5 orders) and represents the limit of the current constant-viscosity, no-Sutherland implementation at high Mach with a thin laminar boundary layer.

5. **Constant viscosity.** No Sutherland law or turbulence model is implemented. All viscous cases use $\mu = \rho_\infty V_\infty L_{\mathrm{ref}}/\mathrm{Re}$ (constant).

6. **Extensibility.** The code is organised with distinct modules for mesh/geometry, residual assembly, boundary conditions, time integration, MPI, and output, providing clear extension points for 3-D, RANS, multi-species, and alternative EOS.